

Sayali Pingle

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EDUCATION

Duke University

Aug 2023 – May 2025

MS in Statistical Science

Durham, NC

- Courses: Predictive Models, Bayesian Statistics, Probabilistic ML, Inferential Statistics, Deep Learning, SQL, NLP
- Teaching Assistant: POLSCI 671- Applied Bayesian Models, STA 101 - Statistical Inference, COMPSCI 370 - Intro to AI

Narsee Monjee Institute of Management Studies (NMIMS)

Jul 2019 – May 2022

BS in Applied Statistics and Analytics

Mumbai, India

- GPA: 3.82/4; Courses: Probability, Linear Algebra, Calculus, Data Mining and Visualization, Regression, Classification

WORK EXPERIENCE

ChessMind AI

Aug 2024 – Present

AI Engineer

Remote, USA

- Built **LLM- Agentic RAG model** with Langgraph and Gemini to give Grandmaster level insight for 1K+ monthly users.
- **Optimized prompts** with a modularized DSPy framework, incorporating Bootstrap Few-shot and chain-of-thought reasoning, resulting in a 35% improvement in response quality. **Implemented testing framework using Pytest.**
- Researched and developed chess-specific evaluation metrics (e.g., strategic depth, threat awareness) and deployed a holistic **evaluation pipeline to assess performance** for LLM response relevance and quality.
- **Analyzed user data** and tailored responses to ELO ratings and play styles, achieving 15% increase in user satisfaction.

Fidelity Investments

Jun 2024 – Aug 2024

Data Science Intern

Durham, NC

- Developed 'Marble Chat'- **Retrieval Augmented Generation (RAG)** ChatBot to **aid in business decisions** and **detect profitable investment options** using LlamaIndex, saving 100+ hours of manual analysis with 0.5 sec of response time.
- Built an **automated data mining pipeline** for 20+ websites for real-time financial data ingestion. **Improved text data extraction accuracy** by 30% using prompt optimization and parser customization to **handle unstructured data.**
- Independently **deployed an end-to-end scalable production pipeline** to the **cloud** using **Microsoft Azure** and Docker.

Pangea Tech Solutions

Jun 2022 – Jun 2023

Data Scientist at Aditya Birla Group

Bengaluru, India

- Engineered **large-scale Supply Chain Analytics dashboard** processing 30K+ rows daily to monitor and forecast operations real-time, **develop metrics to enhance business performance**, saving \$75M in operational costs annually.
- **Streamlined Transport Efficiency** by optimizing truck delivery routes, created algorithms to identify GPS black spots using geospatial data and google API with **PySpark** and DataBricks leading to 10% reduction in carbon emission.
- Implemented **XgBoost** to forecast truck delivery time (92.5% accuracy) reducing mean off-loading wait time by 10 hr.
- **Automated routine tasks** by **creating Python scripts for Data Quality and Data Validation** checks reducing 8 hr work to 10 mins. **Managed clusters and debugged data pipelines** with the Data Engineering team on **Databricks and SQL.**
- **Developed advanced visualizations** in **Tableau**, **performed EDA** to refine key metrics (KPIs) & coordinated with the front-end team to **enhance dashboard UI/UX**, ensuring seamless implementation with well-**documented processes.**
- Fostered collaborative communication & presented data -driven business strategies and recommendations to clients.

SKILLS

- **Languages:** Python, R, SQL, Spark, C, C++, Git
- **Libraries:** Pytorch, LlamaIndex, Langchain, Pandas, Numpy, Streamlit, Bs4, DSPy, Ggplot, RShiny, Pytest, Scipy, Nltk
- **Software:** Databricks, AWS S3, Microsoft Azure, GitHub, Power BI, Tableau, SPSS, PostgreSQL, Snowflake, MySQL

PROJECTS

[Text to Image synthesis](#) of COCO dataset with Multimodal Feature fusion, Duke University

- 90% image-text coherence by **Conditional VAE with CLIP** based text feature extraction and **Gated Fusion** method.

Singapore Traffic Analysis by Object Detection, Duke University

- Created [HuggingFace dataset](#) with 700K+ traffic images; **Performed A/B testing** on truck activity near marine ports to detect alpha signal. Deployed Streamlit dashboard with **time series analysis** for long and short-term traffic patterns.